Computational Biomimetics of Winged Seeds

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——A case of Inverse Design

Winged Seeds（带翅种子or翅果）

种子与气流相互作用 → **长距离、稳定**的被动飞行

微型飞行器、无源飞行器的仿生原型

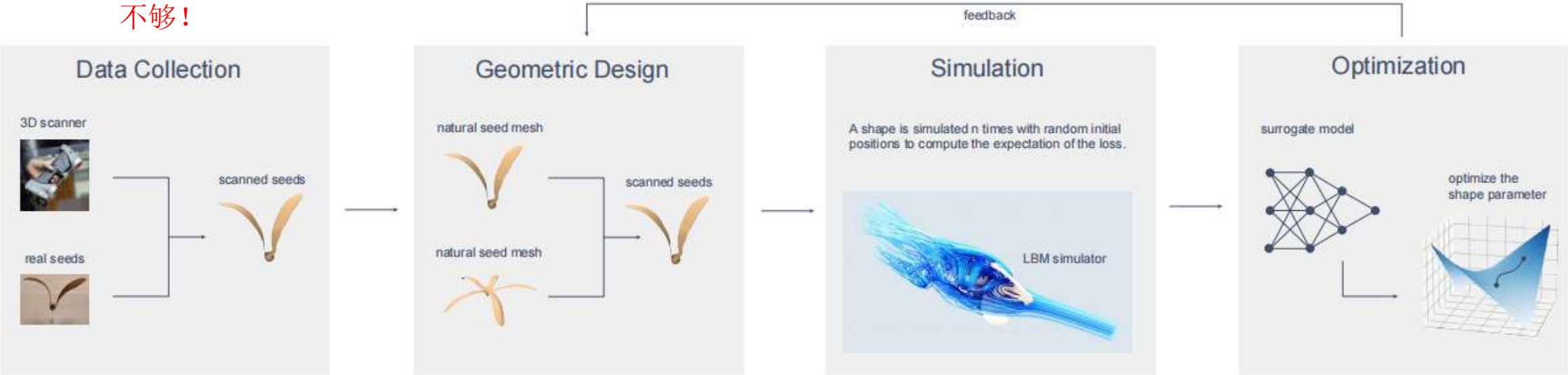
指导设计

如何从中得到启发，设计空气动力学性能优良、甚至比自然界翅果性能更好的形状？

枫树种子

Pipeline

14种翅果数据
不够！



三维扫描
采集真实数据

形状生成

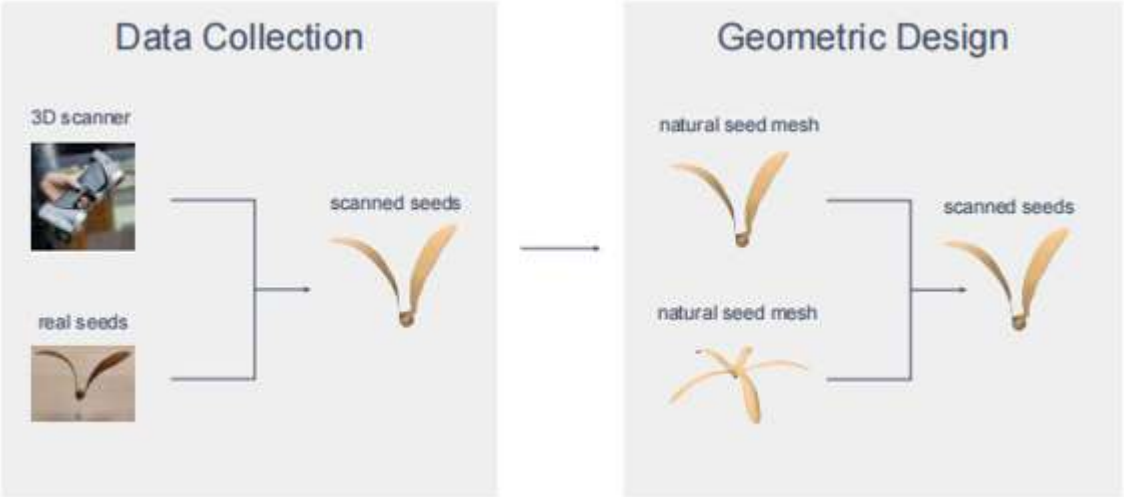
物理仿真

优化

贡献1：收集自然界14种翅果的几何形状模型
贡献2：开发基于微分同胚和测地坐标插值的形状设计方法
贡献3：提出以期望为目标的优化任务，促进优化任务的解决

Pipeline　难点1

14种翅果数据
不够！



三维扫描
采集真实数据

微分同胚和测地坐标插值

李群	李代数
非线性空间	线性空间
李代数本质为李群在单位元的切空间 李代数通过指数映射转化为李群	

微分同胚：$\mathbb{R}^3 \to \mathbb{R}^3$的光滑双射$g$
（天然保证形变无自交、无折叠、保光滑性）

微分同胚群：$\mathbb{R}^3$上所有g构成的集合$\mathcal{G}'$

$\mathcal{G}'$是一个李群，对应李代数$\mathfrak{g}'$（向量场的集合）

例

李群：旋转矩阵的集合

$$\begin{bmatrix} \cos\theta + (1-\cos\theta)x^2 & (1-\cos\theta)xy - (\sin\theta)z & (1-\cos\theta)xz + (\sin\theta)y \\ (1-\cos\theta)yx + (\sin\theta)z & \cos\theta + (1-\cos\theta)y^2 & (1-\cos\theta)yz - (\sin\theta)x \\ (1-\cos\theta)zx - (\sin\theta)y & (1-\cos\theta)zy + (\sin\theta)x & \cos\theta + (1-\cos\theta)z^2 \end{bmatrix}$$

李代数：旋转向量的集合$v = [x, y, z]^T$

$$\begin{bmatrix} 0 & -z\theta & y\theta \\ z\theta & 0 & -x\theta \\ -y\theta & x\theta & 0 \end{bmatrix}$$

一种光滑变换（李群）和该变换的向量化表征（李代数）

Pipeline　难点1

S：离散网格点为$q_1,\dots,q_N$

14种翅果形状：$T^1,T^2,\dots,T^{14}$ ⟶ 假设由统一的初始形状S演化而来 ⟶ 微分同胚$T^k=g^k(S)$

李代数$u^k(\cdot)=\sum_{i=1}^{N}K(q_i,\cdot)p_i^k$　通过有限的动量项p_i^k表征连续向量场　$\mathfrak{g}'\to\mathfrak{g}^*$

记$p^k(0)=\{p_i^k(0)\}_{i=1\ to\ N}$为形状$T_k$的测地线坐标

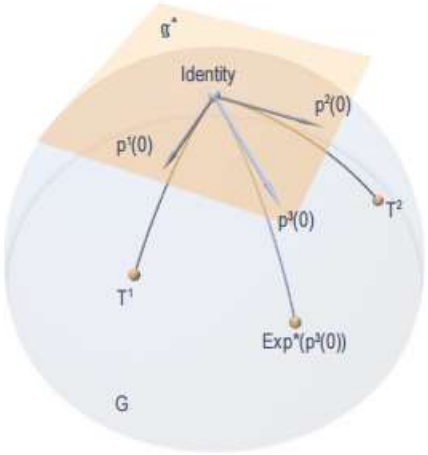
测地流（略去k）

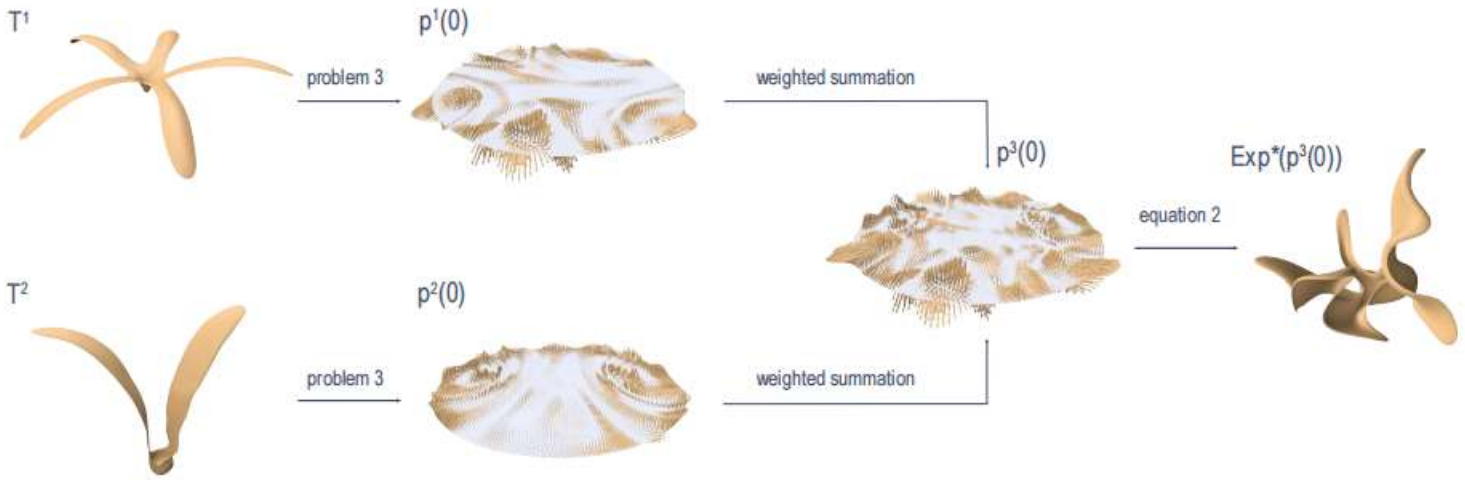
$$\begin{cases}\dfrac{dq_i(t)}{dt}=u(t,q_i(t))\\[2ex]\dfrac{dp_i(t)}{dt}=-\displaystyle\sum_{j=1}^{N}\nabla_1K\big(q_i(t),q_j(t)\big)p_i(t)^Tp_j(t)\end{cases}$$

约束

优化目标

$$\min_{p^1(0)}e\big(Exp^*(p^1(0)),T^1\big)+\frac{1}{2}\sum_{i,j}K\big(q_i(0),q_j(0)\big)p_i^1(0)^Tp_j^1(0)$$





Pipeline　难点1

14组$p^k(0) = \{p_i^k(0)\}_{i=1\ to\ N}$测地线坐标 ⟶ 插值$p^{new}(0) = \sum_{k=1}^{14} w_k p^k(0)$ ⟶ 光滑新形状

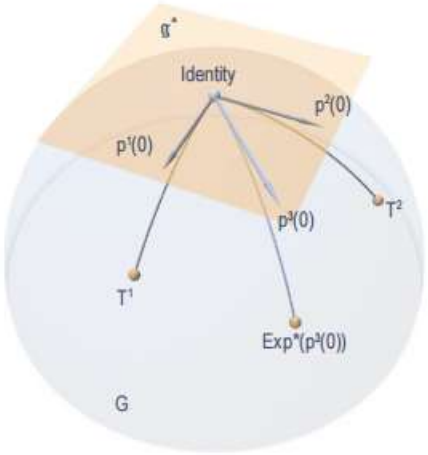
非线性光滑变换

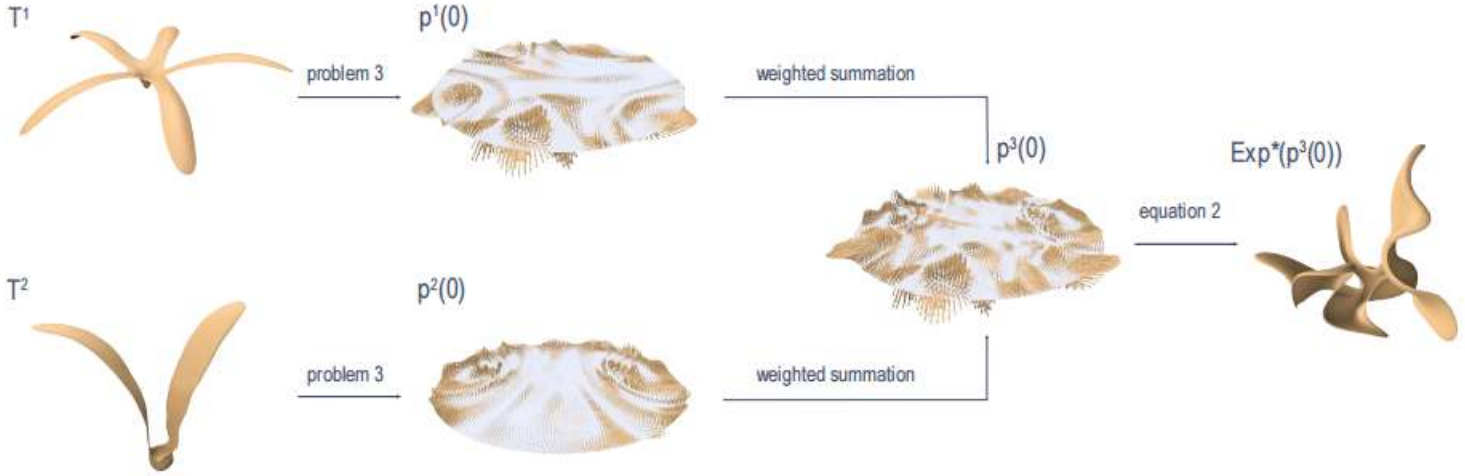
↓

向量表征

↓

插值，生成形状

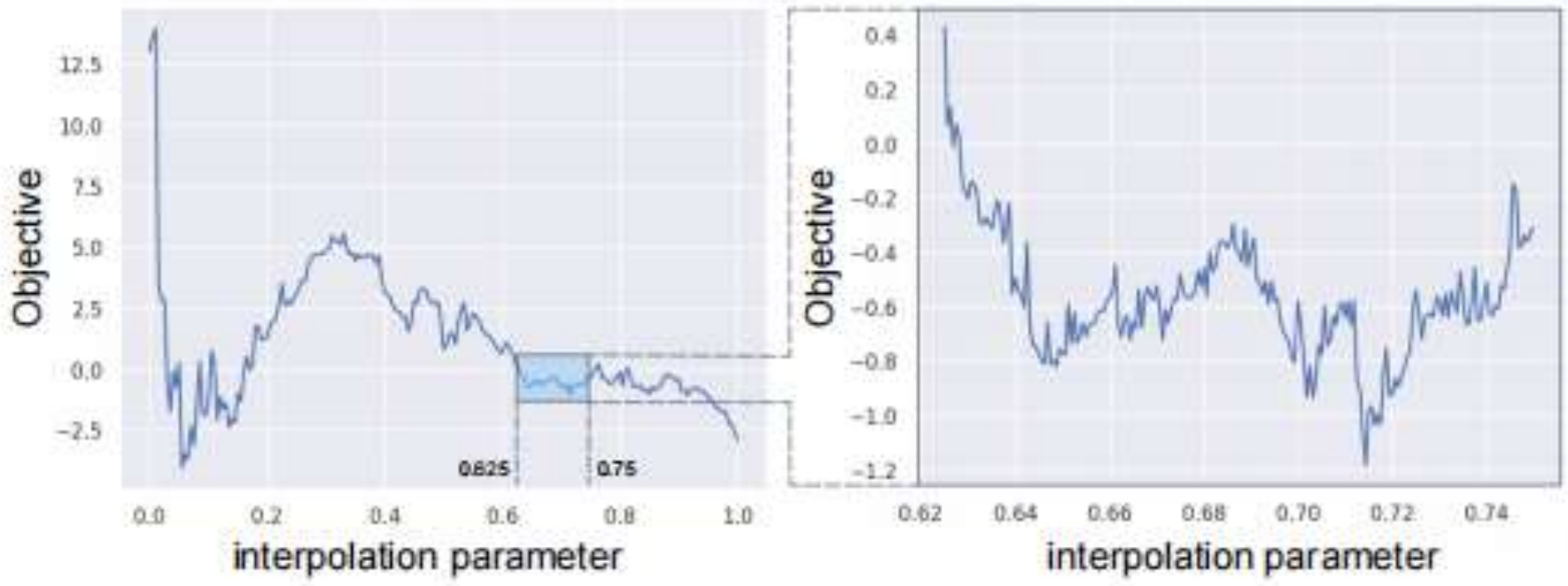




Pipeline　难点2

目标函数$f(s,\alpha)$， s为形状，α为初始条件， f表示具体任务（如飞行距离、下落高度）

$f(s,\alpha)$对α极度敏感，目标函数高度混沌，无显式梯度，有限差分法估计梯度会被噪声覆盖



Pipeline　难点2

$f(s,\alpha)$对α极度敏感，目标函数高度混沌 ⟶ 用代理模型$h(s,\theta)$局部拟合期望

目标函数期望

$$\min_s \mathbb{E}_{\alpha\sim p}[f(s,\alpha)] = \min_s \int p(\alpha) f(s,\alpha) d\alpha$$

局部拟合损失函数（s_i附近）

$$l(s_i,\theta) = \int \kappa(s-s_i)\left(\mathbb{E}_{\alpha\sim p}[f(s,\alpha)] - h(s,\theta)\right)^2 ds$$

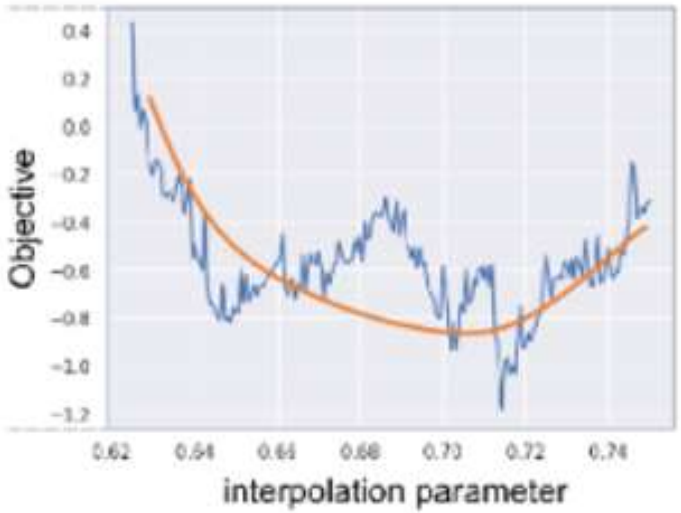
$$\Leftrightarrow l(s_i,\theta) = \int \kappa(s-s_i)\mathbb{E}_{\alpha\sim p}\left[\left(f(s,\alpha) - h(s,\theta)\right)^2\right] ds$$

$$\frac{\partial l}{\partial \theta} \approx \frac{\partial}{\partial \theta}\frac{1}{N}\sum_{j=1}^{N}\left(h(s_j,\theta) - f(s,\alpha)\right)^2$$

$$s_j \sim \kappa(s-s_i), \alpha \sim p(\alpha)$$

可以直接用蒙特卡洛采样替代期望计算
代码实现只需计算MSE

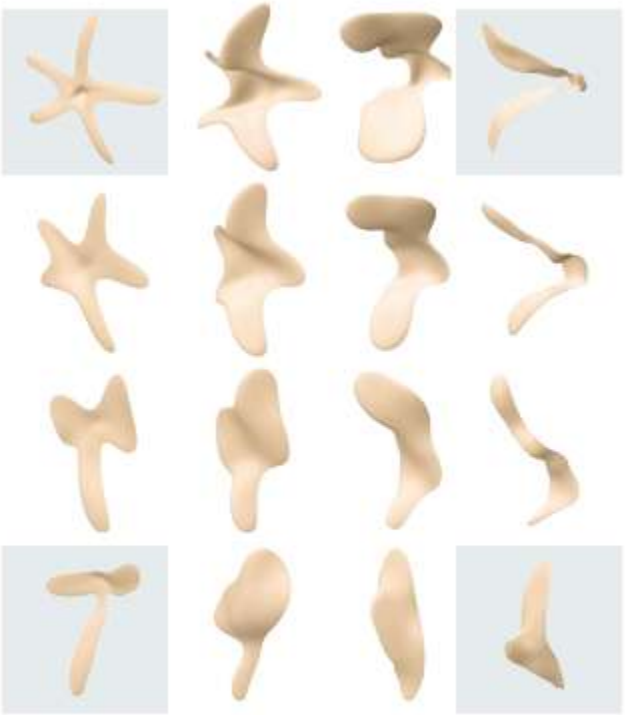
1. Initialize: s_i
2. Sampling: N组 $s_j \sim \kappa(s-s_i), \alpha_j \sim p(\alpha)$
3. Simulation: 黑盒仿真 $f(s_j,\alpha_j)$
4. Training: $h(s,\theta)$ 针对s_i，使用左式
5. Gradient: $\nabla h(s,\theta)$，更新s_{i+1}
6. Loop: go to Step2 until convergence



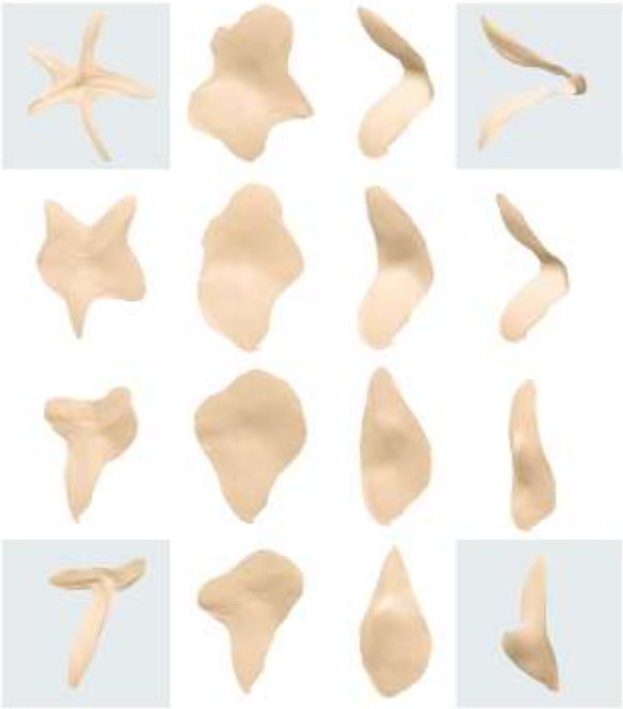
局部代理拟合
↓
代理梯度下降

Experiments

形状插值



Geodesic Shooting

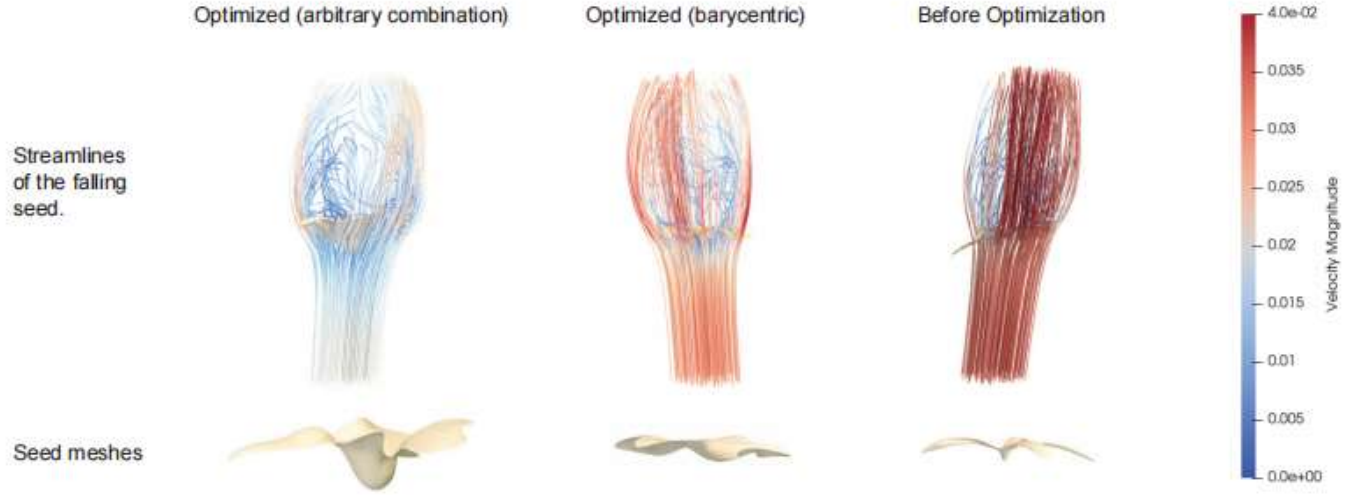


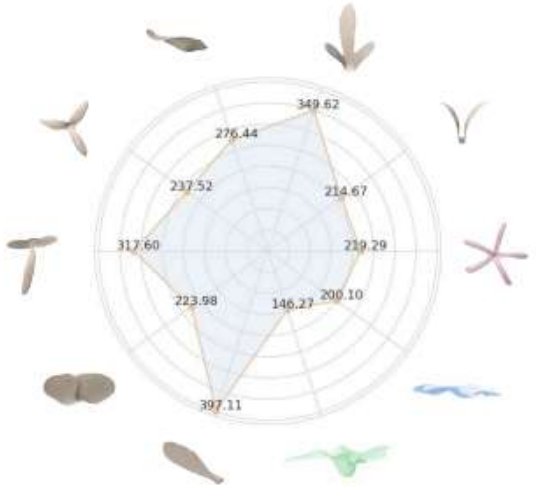
MeshPooling

UDF encoder-decoder

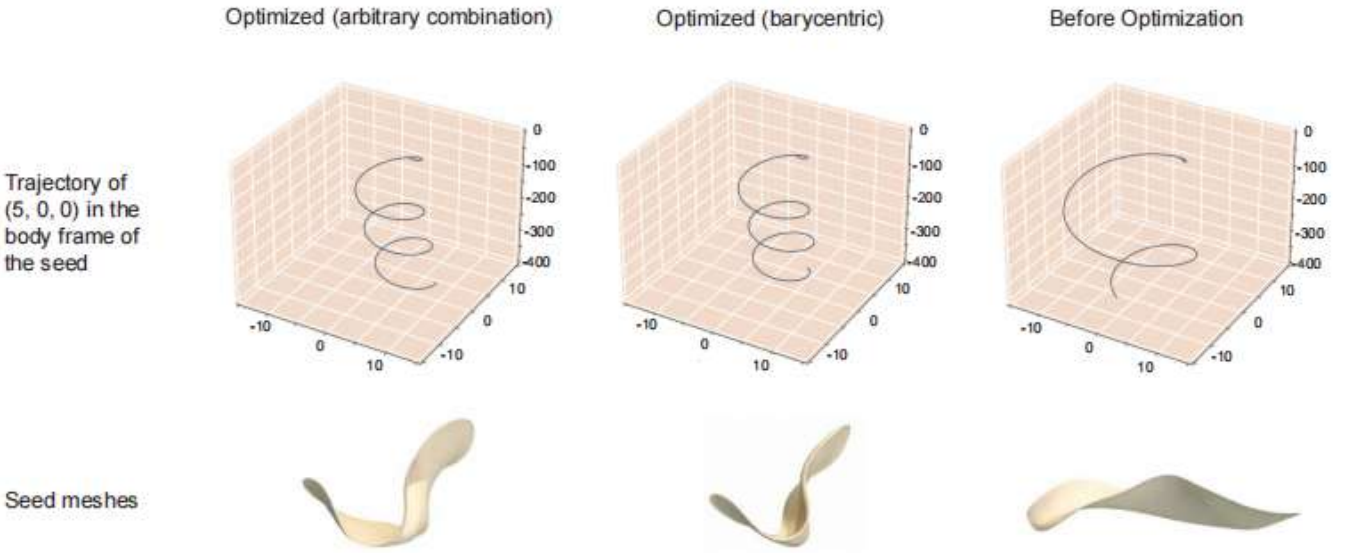
Experiments

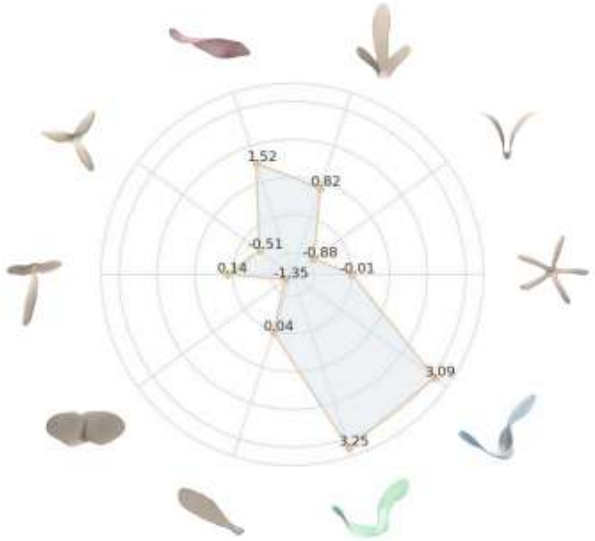
坠落实验





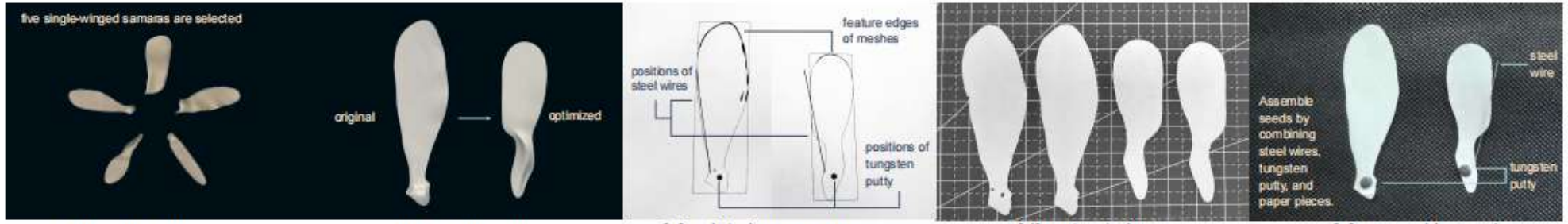
旋转实验



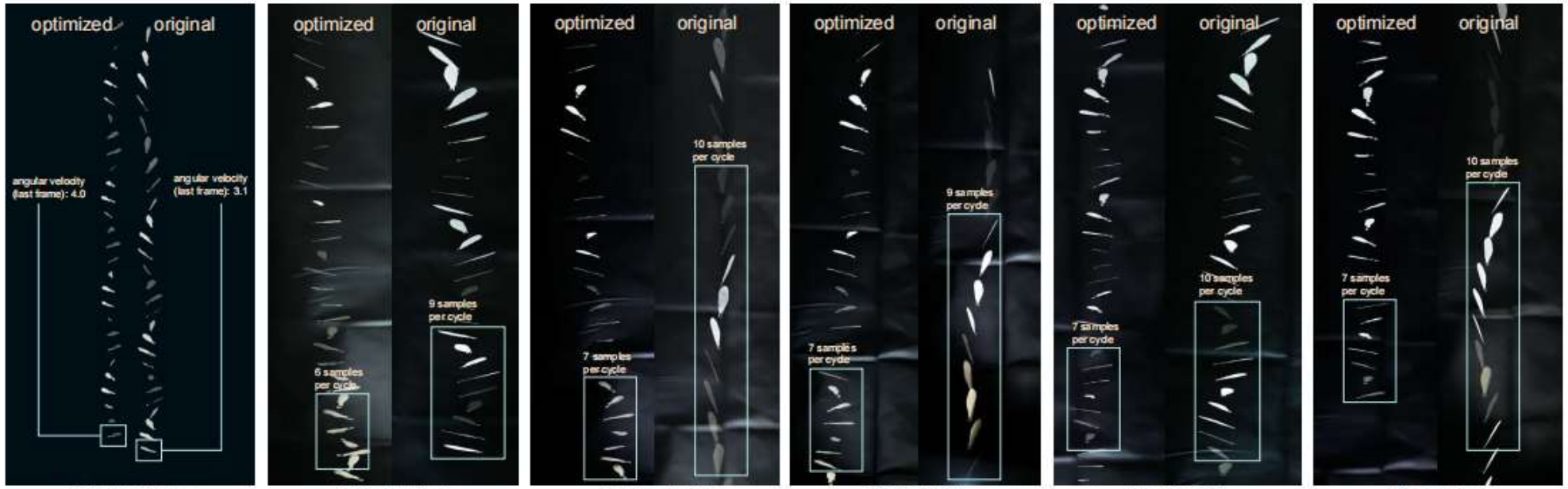


Experiments

真实实验



[a]: shape space [b]: optimization [c]: print shapes on paper [d]: cut out shapes [e]: assemble seeds



[a]: simulation rotational speed +29.0%

[b]: 1st trial rotational speed +50.0%

[c]: 2nd trial rotational speed +42.9%

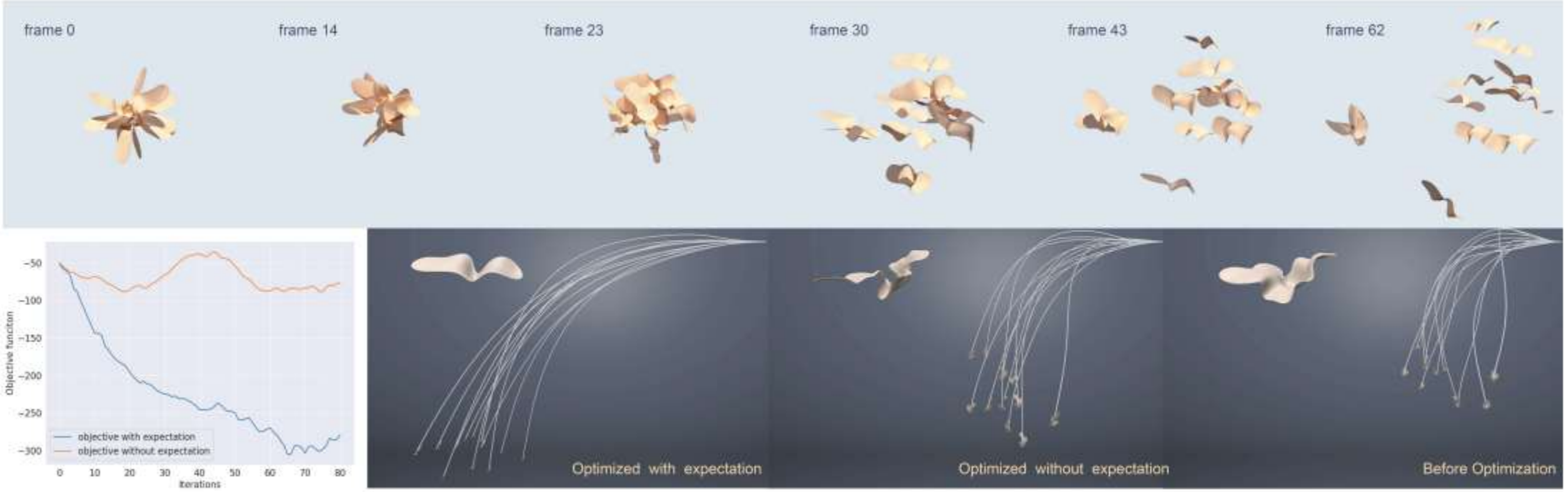
[d]: 3rd trial rotational speed +28.6%

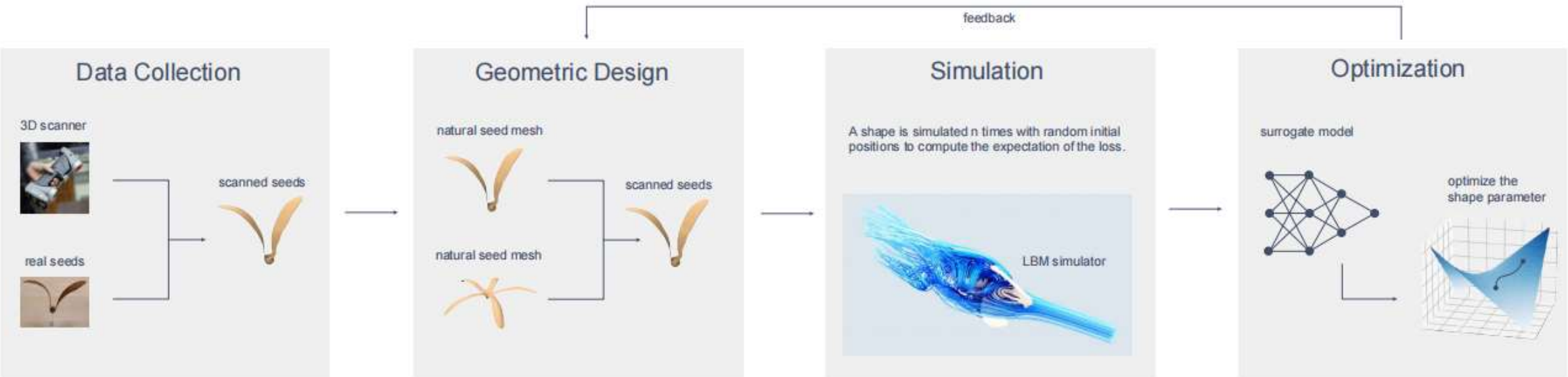
[e]: 4th trial rotational speed +42.9%

[f]: 5th trial rotational speed +42.9%

Experiments

期望优化&原始优化对比





数据　　数据表征、参数化　　物理仿真　　优化

可能可以深入研究的：李群、李代数；哈密顿动力学

THANKS

$$l(s_i,\theta) = \int \kappa(s - s_i)\left(\mathbb{E}_{\alpha\sim p}[f(s,\alpha)] - h(s,\theta)\right)^2 ds$$

$f(s,\alpha)$方差不影响梯度

$$\Leftrightarrow l(s_i,\theta) = \int \kappa(s - s_i)\mathbb{E}_{\alpha\sim p}\left[\left(f(s,\alpha) - h(s,\theta)\right)^2\right]ds$$

$$= \int\int \kappa(s - s_i)p(\alpha)\left(f(s,\alpha) - h(s,\theta)\right)^2 d\alpha ds$$

$$\frac{dl}{d\theta} = \int\int 2\kappa(s - s_i)p(\alpha)\frac{\partial h(s,\theta)}{\partial\theta}\left(h(s,\theta) - f(s,\alpha)\right)d\alpha ds$$

$$\approx \frac{1}{N}\sum_j \frac{2\kappa(s_j - s_i)p(\alpha_j)}{p(\alpha_j,s_j)}\frac{\partial h(s_j,\theta)}{\partial\theta}\left(h(s_j,\theta) - f(s_j,\alpha_j)\right)$$

$$= \frac{\partial}{\partial\theta}\frac{1}{N}\sum_j \frac{\kappa(s_j - s_i)p(\alpha_j)}{p(\alpha_j,s_j)}\left(h(s_j,\theta) - f(s_j,\alpha_j)\right)^2$$

$$= \frac{\partial}{\partial\theta}\frac{1}{N}\sum_j \left(h(s_j,\theta) - f(s_j,\alpha_j)\right)^2$$